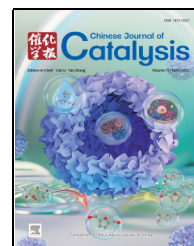


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Article

Catalytic upgrading of waste PET to dimethyl cyclohexane-1,4-dicarboxylate over defective sulfonated UiO-66_{def}-SO₃H supported Ru catalyst

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ABSTRACT

Poly(ethylene glycol-co-1,4-cyclohexanedimethanol terephthalate) (PETG) possesses excellent properties and stability than traditional poly(ethylene terephthalate) (PET). However, the production and application of PETG are restricted by the expensive monomer (1,4-cyclohexanedimethanol, CHDM). Direct upgrading of waste PET to dimethyl cyclohexane-1,4-dicarboxylate (DMCD) can promote the production of CHDM in large scale. In this work, a bifunctional Ru/UiO-66_{def}-SO₃H catalyst was synthesized and utilized in coupled methanolysis (of waste PET to dimethyl terephthalate (DMT)) and hydrogenation (of DMT to DMCD) under mild condition. Characterizations revealed that Ru/UiO-66_{def}-SO₃H possessed mesopores (dominant channels of 2.72 and 3.44 nm), enlarged surface area (998 m²·g⁻¹), enhanced acidity (580 μmol·g⁻¹), and Ru nanoparticles (NPs) dispersed highly (45.1%) compared to those of Ru/UiO-66. These combined advantages could accelerate the methanolysis and hydrogenation reactions simultaneously, promoting the performance of direct upgrading of PET to DMCD in one pot. In particular, the conversion of PET and yield of DMCD over Ru/UiO-66_{def}-SO₃H reached 100% and 97.7% at 170 °C and 3 MPa H₂ within 6 h. Moreover, Ru/UiO-66_{def}-SO₃H was also capable for the upcycling of waste PET-based products including beverage bottles, textile fiber and packaging film to DMCD.

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1. Introduction

Poly(ethylene terephthalate) (PET) has good physical and chemical properties, which is widely used in bottles, fiber and film [1–3]. However, the transparency, ultraviolet shielding, heat resistance and gas barrier properties of traditional PET products are insufficient [4,5]. It was reported that these properties could be achieved by incorporating co-monomer with specific chemical structures, such as 1,4-cyclohexanedimetha-

nol (CHDM) [6,7].

Poly(ethylene glycol-co-1,4-cyclohexanedimethanol terephthalate) (PETG) and poly(1,4-cyclohexylenedimethylene terephthalate glycol) (PCTG) were recommended as alternates of PET since 1980s [8,9], and PETG and PCTG could be synthesized *via* esterification and/or co-polycondensation of *p*-phthalic acid (PTA), ethylene glycol, and CHDM [10,11]. After the addition of CHDM, PETG and PCTG show low coloration, good chemical resistance and high toughness [12,13]. Unfortu-

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nately, the real production and application of PETG or PCTG are restricted by the high-cost and limited productivity of CHDM [14,15].

Previously, CHDM was synthesized from dimethyl terephthalate (DMT) under harsh conditions using petroleum-based PTA as feedstock in several steps (see routine (a) in Scheme 1) [16,17]. In which, DMT was obtained from PTA in supercritical methanol (at 280–300 °C and 20–40 MPa) [18]. After that, DMT was hydrogenated to dimethyl cyclohexane-1,4-dicarboxylate (DMCD) under harsh conditions (250 °C and 30 MPa H₂), and followed hydrogenation of DMCD to CHDM at 250 °C and 38 MPa H₂ [19].

Recently, it was reported that waste PET or PET-based products could be utilized as the feedstock for the synthesis of DMT [20,21]. And then, DMT could be hydrogenated to DMCD and CHDM efficiently under mild conditions (see routine (b) in Scheme 1) [22–25]. This technology was safer and more economical which attracted more attention for the upcycling of waste PET. However, the separation and purification of formed DMT from solid catalyst and/or remaining PET are quite difficult since the poor solubility (~1.5 wt% in methanol at 25 °C) and high boiling point of DMT (284 °C) [26,27], and a large amount of poisonous trichloromethane is added to get DMT separated [21].

To overcome this drawback, direct upgrading of PET to DMCD instead of DMT is of great significance [28,29]. As the boiling point of DMCD (~260 °C) is low and its solubility (> 30 wt% in methanol at 25 °C) is high, the formed DMCD can be separated and purified easily compared to DMT [30]. At the same time, it was estimated that the cost of equipment and energy consumption of direct formation of DMCD from waste PET would decrease significantly compared to the traditional two-step conversion route (Tables S1–S3), and the cost of DMCD from one-step routine was about 26% lower than that of the two-step process [31]. But, the coupled methanolysis reaction (of PET to DMT) and hydrogenation reaction (of DMT to DMCD) under mild conditions over a single solid catalyst (see routine (c) in Scheme 1) is quite difficult. Most recently, Mo et al. found that a 0.5Pd5Ni/CeO₂ catalyst could give an excep-

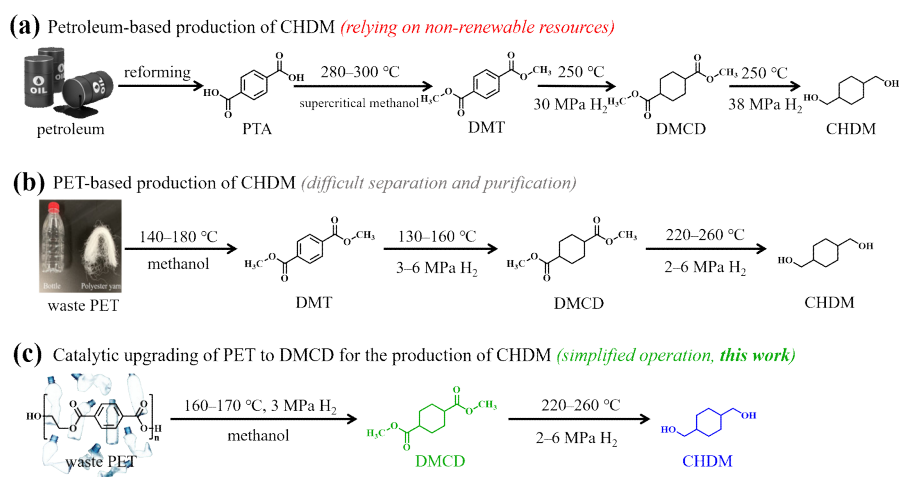
tional yield of DMCD (86.5%) with high selectivity (97.4%) at 180 °C within 10 h (with 0.1 g PET, 0.1 g 0.5Pd5Ni/CeO₂, 5 g CH₃OH and 6 MPa H₂ in feed) [32]. They also found that the production of CHDM *via* direct conversion of PET to DMCD had higher benefits than traditional routines.

In this submission, a defective sulfonated UiO-66_{def}-SO₃H was synthesized *via* replacing partial 1,4-benzenedicarboxylic acid (H₂BDC) with 2-mercaptobenzoic acid (MBA), and the formed UiO-66_{def}-SO₃H was further utilized as the support of Ru nanoparticles (NPs). And we hope the mesoporous channels, enlarged surface area and strong acidity of defective sulfonated UiO-66_{def}-SO₃H can accelerate the methanolysis step (of waste PET to DMT) and increase the dispersion of Ru, and these combined advantages may further promote its performance in direct upgrading of waste PET to DMCD in one pot. The structure and properties of prepared Ru/UiO-66_{def}-SO₃H catalyst were characterized with varied techniques, and its performance in direct upgrading of waste PET to DMCD was discussed.

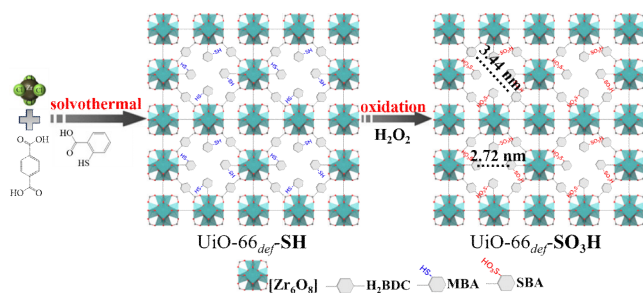
2. Experimental

2.1. Catalyst preparation

All materials and chemicals used in this work were addressed in Supporting Information. Defective sulfonated UiO-66_{def}-SO₃H was synthesized *via* solvothermal method according to the published literature in Scheme 2 [33]. At first, 0.700 g ZrCl₄, 0.250 g H₂BDC and 0.096 g MBA were dissolved in 50 mL *N,N'*-dimethylformamide (DMF) under vigorous stirring for 30 min. Then, above mixture was transferred to a Teflon-sealed autoclave and reacted at 140 °C for 24 h. Subsequently, the yellow solid (denoted as UiO-66_{def}-SH) was collected *via* centrifugation, washed with ethanol for five times and dried in vacuum at 80 °C for 12 h. Afterwards, 0.500 g UiO-66_{def}-SH was dispersed in 20 mL ethanol and sulfonated with 10 mL 30 wt% H₂O₂ (aq) under stirring at room temperature for 12 h. Finally, the white solid (denoted as UiO-66_{def}-SO₃H) was obtained using centrifugation, washed



Scheme 1. Different routines for the synthesis of CHDM.



Scheme 2. Schematic illustration for the preparation of UiO-66-def-SO₃H.

with ethanol for five times and dried in vacuum at 80 °C for 12 h. Perfect UiO-66 was synthesized using H₂BDC only as ligand in the similar procedure as above, which was described in detail in Supporting Information (Scheme S1).

Ru/UiO-66-def-SO₃H was prepared *via* impregnation (Scheme S2(a)). Typically, 0.500 g UiO-66-def-SO₃H was dispersed in 10 mL deionized water and pretreated in ultrasonic bath for 30 min. Then, RuCl₃ aqueous solution (2.5 mL, 0.01 g(Ru)·mL⁻¹) was added dropwise to the above mixture. After stirring at room temperature for 8 h, the black solid was collected *via* evaporating the suspension. At last, the solid was reduced at 170 °C under H₂ flow (30 mL·min⁻¹) for 1 h to obtain Ru/UiO-66-def-SO₃H. Besides, the preparation of Ru/UiO-66 was similar as that of Ru/UiO-66-def-SO₃H, except replacing UiO-66-def-SO₃H with UiO-66 (Scheme S2(b)).

The structure, composition and properties of defective sulfonated UiO-66-def-SO₃H and Ru/UiO-66-def-SO₃H catalyst were characterized using powder X-ray diffraction (XRD), inductively coupled plasma atomic emission spectroscopy (ICP-AES), Fourier transform infrared spectroscopy (FTIR), static water contact angle test, N₂ adsorption-desorption measurement, thermogravimetric and differential thermal analysis (TG-DTA), H₂ temperature-programmed desorption test (H₂-TPD), acid-base titration, pyridine adsorption infrared spectra (pyridine-IR), scanning electron microscopy (SEM), transmission electron microscopy (TEM), elemental energy dispersive spectroscopy (EDS), and X-ray photoelectron spectroscopy (XPS). The procedures of these characterization techniques were introduced in *Supporting Information* in detail.

2.2. Catalyzed reaction

Direct upgrading of PET to DMCD was carried out in a stainless-steel autoclave with a Teflon inner layer (50 mL). In a typical run, 0.1 g catalyst, 2.0 g PET, 20 mL methanol and a magnetic stirrer were added to the autoclave. The autoclave was sealed and flushed with purified H₂ to remove air thoroughly. Next, the autoclave was pressurized to the desired pressure, and placed in a heating jacket. As being preheated to the required temperature, the stirrer was started at 1000 r·min⁻¹. After reacting for a certain time, the reactor was cooled to room temperature, and the reaction mixture was dissolved with exceed trichloromethane and transferred into a volumetric flask (50 mL). The mixture was spun down and precipitated to separate reaction liquid with remained solid. The products in

liquid phase were analyzed using a gas chromatograph (Shimadzu, 14B) equipped with a 30-m capillary column (HP-5MS) and a flame ionization detector and confirmed by gas chromatography-mass spectrometry (Agilent 6890-5973 N), whereas the remained solid was washed with ethanol, dried and weighed to calculate the degradation rate of PET (see *Supporting Information*). The conversion of PET and yield of DMCD were calculated as follows.

$$\text{Conversion of PET} = \frac{\text{mole of benzene ring in added PET} - \text{mole of benzene ring in remained PET}}{\text{mole of benzene ring in added PET}} \times 100\% \quad (1)$$

$$\text{Yield of DMCD} = \frac{\text{mole of formed DMCD}}{\text{mole of benzene ring in added PET}} \times 100\% \quad (2)$$

The performance of the recycled catalyst was examined in the same procedures as above. After the reaction, trichloromethane was added to the reaction mixture to dissolve the products. The leftover catalyst was collected *via* centrifuged, washed with ethanol, dried at 80 °C in vacuum for 12 h, and weighed before next recycle.

The procedures of separated methanolysis of PET and hydrogenation of DMT, coupled methanolysis and hydrogenation of PET in one pot were described in detail in *Supporting Information*.

3. Results and discussion

3.1. Catalyst characterization

XRD patterns of UiO-66 and UiO-66-def-SO₃H found that the diffraction patterns of UiO-66 and UiO-66-def-SO₃H were consistent with those of UiO-66 (Fig. S1) [34], indicating the successful synthesis of UiO-66 and the structure preservation in UiO-66-def-SO₃H. However, the diffraction intensity of UiO-66-def-SO₃H decreased slightly compared to that of UiO-66 because of the formation of defects. Meanwhile, two weak diffraction peaks appeared at $2\theta = 3^\circ - 7^\circ$ in UiO-66-def-SO₃H, which were absent in UiO-66. These results implied the formation of clusters [Zr₆(OH)₄O₄] defect along with ligands substitution in UiO-66-def-SO₃H [33,35].

After the loading of Ru, the characteristic peaks of UiO-66 remained stable in both samples, and lattice parameters ($a = b = c$) of Ru/UiO-66 (2.07476 nm) and Ru/UiO-66-def-SO₃H (2.07404 nm) were close to that of UiO-66 (2.07465 nm, Fig. 1 and Table 1) [33]. Likewise, the diffraction of clusters [Zr₆(OH)₄O₄] defect ($3^\circ - 7^\circ$) remained in Ru/UiO-66-def-SO₃H. These results indicated that the introduction of Ru had little influence on the structure of UiO-66 and UiO-66-def-SO₃H. The content of Ru in Ru/UiO-66 and Ru/UiO-66-def-SO₃H was 4.79 and 4.87 wt%, respectively (Table 1), but no characteristic peaks of metallic Ru (JCPDS No. 01-070-0274) was detected in either (Fig. 1), inferring the high dispersion of Ru and/or the poor crystallinity of Ru [36].

FTIR analysis found that the strong signals of symmetric and asymmetric stretching vibration of Zr-O in Ru/UiO-66 and Ru/UiO-66-def-SO₃H nodes appeared at 755 and 670 cm⁻¹, respectively (Fig. S2). Moreover, the typical peaks at 1580, 1500

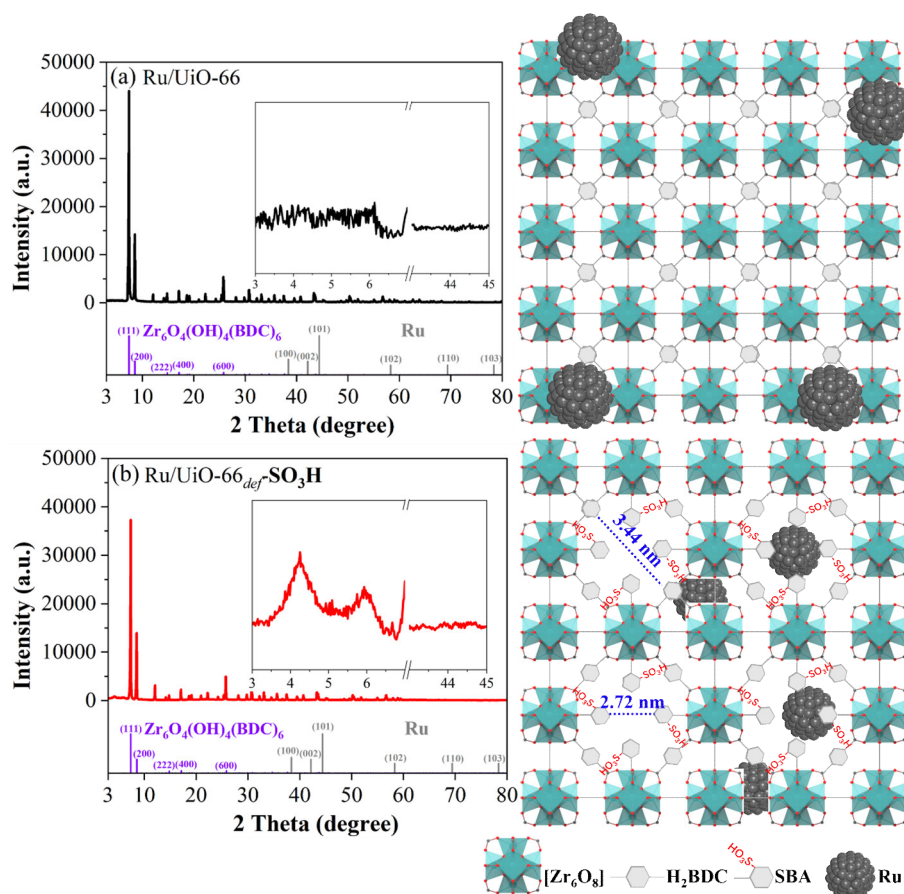


Fig. 1. XRD patterns including enlarged diffractions at 3°–7°, 43°–45° and structural representation of Ru/UiO-66 (a) and Ru/UiO-66_{def}-SO₃H (b).

and 1410 cm⁻¹ were attributed to the stretching vibration of the benzene ring from BDC and/or MBA, and the intensity of bands at 1660 cm⁻¹ belonged to the stretching vibration of C=O in carboxylate [37]. These results confirmed the framework structure of UiO-66 remained stable in both catalysts. As for Ru/UiO-66_{def}-SO₃H, the asymmetric and symmetric stretching vibration of O=S=O appeared at 1285 and 1220 cm⁻¹, while the peaks at 1040 and 630 cm⁻¹ corresponded to the stretching vibration of C–S and S–O, respectively [38], which indicated the successful introduction of –SH and the further oxidation to –SO₃H (Scheme S3). In addition, the peak at 3440 cm⁻¹ was accredited to the hydroxyl in ligands and/or absorbed water [39]. The hydroxyl peak in Ru/UiO-66_{def}-SO₃H was more obvious than that in Ru/UiO-66, inferring that the hydrophilicity of defective sulfonated Ru/UiO-66_{def}-SO₃H was stronger owing to extra sulfonic acid groups and the exposure of unsaturated Zr–O nodes [40].

The static water contact angle of Ru/UiO-66 was 37.4°, and this value decreased to 22.9° in Ru/UiO-66_{def}-SO₃H (Fig. S3), which was the direct evidence that Ru/UiO-66_{def}-SO₃H was more hydrophilic (or less hydrophobic) than Ru/UiO-66. The increasing hydrophilicity of Ru/UiO-66_{def}-SO₃H could enhance the adsorption of PET and its oligomers with terminal hydroxyl, as well as promote the desorption of hydrophobic DMT to accelerate the methanolysis of PET [41].

N₂ adsorption-desorption isotherms and pore size distributions found that Ru/UiO-66 showed type I isotherms with ob-

vious uptake of N₂ at low p/p_0 , indicating the domination of micropores (0.79 and 1.17 nm) as similar as UiO-66 (Fig. S4). And the surface area and microporous volume of Ru/UiO-66 (886 m²·g⁻¹ and 0.48 cm³·g⁻¹, Table 2) decreased slightly than that of UiO-66 (1012 m²·g⁻¹ and 0.57 cm³·g⁻¹), which implied that Ru NPs mainly located on the surface of UiO-66 (Scheme S2(b)).

On the other hand, Ru/UiO-66_{def}-SO₃H presented type IV isotherms with a conspicuous hysteresis loop at relative pressure of 0.5–0.9 (Fig. 2), suggesting newly emerging mesopores (with the diameters of 2.72 and 3.44 nm) in UiO-66_{def}-SO₃H (see Scheme 2). The newly formed mesopores were attributed to that replacing H₂BDC with monocarboxylic acid (MBA) resulted in clusters defect as the partial missing of [Zr₆(OH)₄O₄] [42]. Moreover, the surface area and mesoporous volume of Ru/UiO-66_{def}-SO₃H reduced obviously from 1148 m²·g⁻¹ and

Table 1
The properties of Ru/UiO-66 and Ru/UiO-66_{def}-SO₃H.

Catalyst	Lattice parameter ^a ($a = b = c$, nm)	Ru Content (%)		Ru dispersion ^d (%)
		Bulk ^b	Surface ^c	
Ru/UiO-66	2.07476	4.79	6.13	37.4
Ru/UiO-66 _{def} -SO ₃	2.07404	4.87	4.69	45.1

^a Calculated from the result of XRD.

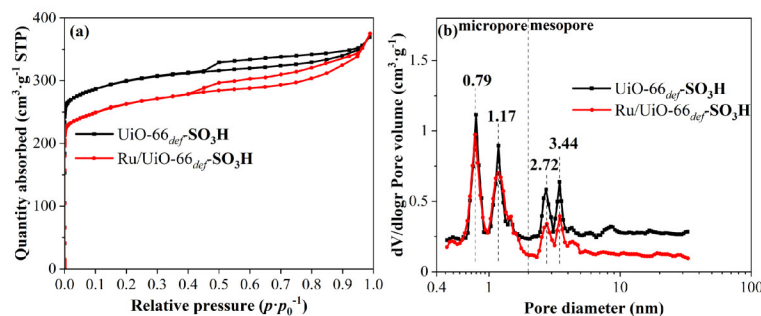
^b Determined using ICP-AES.

^c Calculated by analyzing XPS result.

^d Calculated by $2 \times A \times 10^{-6} \times M_{\text{Ru}}/\omega$, where A and ω represent H₂ absorption and Ru bulk content.

Table 2Textural properties of Ru/Uio-66, Ru/Uio-66_{def}-SO₃H and their supports.

Sample	A_{BET}^a (m ² ·g ⁻¹)	A_{ext}^b (m ² ·g ⁻¹)	A_{int}^b (m ² ·g ⁻¹)	V_{micro}^b (cm ³ ·g ⁻¹)	V_{meso}^c (cm ³ ·g ⁻¹)	V_{pore}^c (cm ³ ·g ⁻¹)
UiO-66	1012	298	714	0.57	0.15	0.72
Ru/Uio-66	886	276	590	0.48	0.13	0.61
UiO-66 _{def} -SO ₃ H	1148	517	631	0.43	0.54	0.97
Ru/Uio-66 _{def} -SO ₃ H	998	453	545	0.39	0.44	0.83

^a Calculated by Brunauer-Emmett-Teller (BET) method.^b Measured by t-plot method.^c Measured by Barrett-Joyner-Halenda (BJH) method.**Fig. 2.** N₂ adsorption-desorption isotherms (a) and pore size distributions (b) of UiO-66_{def}-SO₃H and Ru/Uio-66_{def}-SO₃H.

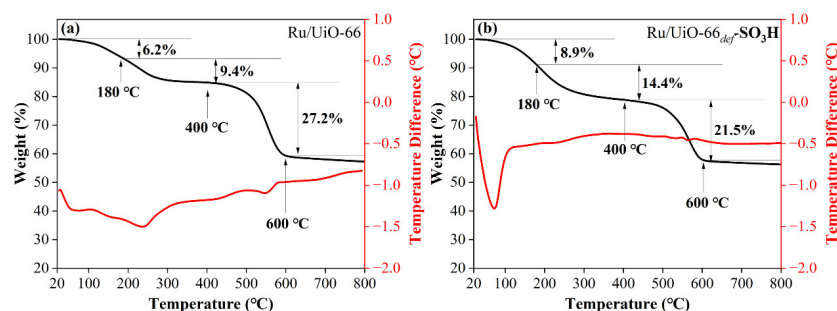
0.54 cm³·g⁻¹ (of UiO-66_{def}-SO₃H) to 998 m²·g⁻¹ and 0.44 cm³·g⁻¹ (Table 2), respectively, which inferred that Ru NPs mainly located inside mesopores (Scheme S2(a)). The emerging mesopores in Ru/Uio-66_{def}-SO₃H would promote the diffusion of waste PET degraded chemicals as well as the contact between reactants and Ru NPs compared to microporous Ru/Uio-66 [43].

Three mass loss stages could be detected during the thermal analysis of Ru/Uio-66 and Ru/Uio-66_{def}-SO₃H under N₂ flow (Fig. 3). At first, the mass loss between 25 and 180 °C might be ascribed to the volatilization of adsorbed solvent and/or water. The second stage from 180 to 400 °C could be attributed to the dehydration of [Zr₆O₄(OH)₄] into [Zr₆O₆] [44]. After that, a significant mass loss in the range of 400–600 °C was observed owing to the decomposition of ligands (H₂BDC and/or MBA) [37]. Finally, the curve altered faintly further raising temperature (> 600 °C). It was noticed that the mass loss at 400–600 °C for Ru/Uio-66_{def}-SO₃H (21.5%) was lower than that of Ru/Uio-66 (27.2%) because partial ligands H₂BDC was replaced by MBA [32]. The mass loss trend of UiO-66 and UiO-66_{def}-SO₃H was similar with that of Ru/Uio-66 and Ru/Uio-66_{def}-SO₃H, indicating that the introduction of Ru had almost no effect on the thermal stability of catalysts (Fig. S5).

Hydrogen adsorption and activation over Ru/Uio-66 and

Ru/Uio-66_{def}-SO₃H were checked via H₂-TPD (Fig. S6). There was mainly one desorption peak centered at 160–240 °C in both catalysts, which was ascribed to H₂ adsorbed on metallic Ru NPs [22]. It was worth noting that the amount of adsorbed H₂ over Ru/Uio-66_{def}-SO₃H (108.8 μmol·g⁻¹) was higher than that over Ru/Uio-66 (88.7 μmol·g⁻¹), as well as the desorption center temperature (200 °C for Ru/Uio-66_{def}-SO₃H while 190 °C for Ru/Uio-66). These results indicated that Ru/Uio-66_{def}-SO₃H processed excellent ability for the adsorption and activation of hydrogen, and further inferred that Ru NPs were highly dispersed on defective sulfonated UiO-66_{def}-SO₃H (45.1%) than that on UiO-66 (37.4%, Table 1).

According to acid-base titration, the acidity of Ru/Uio-66_{def}-SO₃H increased obviously from 290 μmol·g⁻¹ (of Ru/Uio-66) to 580 μmol·g⁻¹, owing to the exposure of Zr–O nodes and the introduction of –SO₃H (Table S4) [32]. Pyridine-IR spectra of Ru/Uio-66 and Ru/Uio-66_{def}-SO₃H exhibited that the typical peaks at 1540 and 1450 cm⁻¹ were corresponding to pyridine adsorbed on Brønsted (Zr–OH and/or –SO₃H) and Lewis (Zr–O nodes) acid sites (Fig. 4(a)). It could be observed that both amounts of Brønsted and Lewis acid sites in Ru/Uio-66_{def}-SO₃H (405 and 175 μmol·g⁻¹) increased compared to those of Ru/Uio-66 (190 and 100 μmol·g⁻¹), due to the introduction of –SO₃H and clusters defects (Table S4). And the

**Fig. 3.** Thermal analysis of Ru/Uio-66 (a) and Ru/Uio-66_{def}-SO₃H (b) in N₂ flow.

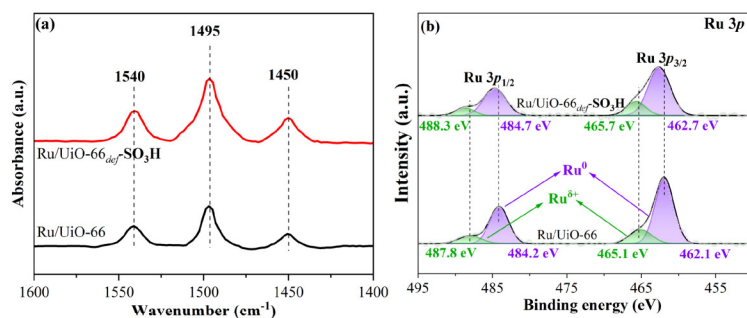


Fig. 4. Pyridine-IR spectra (a) and binding energy of Ru 3p (b) of Ru/UiO-66 and Ru/UiO-66_{def}-SO₃H.

increase of Brønsted acid sites could serve as metal anchoring and promote the dispersion of Ru NPs [45].

The octahedral shaped UiO-66 and UiO-66_{def}-SO₃H were clearly observed in SEM analysis (Fig. S7) [32,37]. High-resolution TEM analysis further disclosed that Ru NPs were evenly dispersed in UiO-66_{def}-SO₃H with 2.5 nm of the mean size (Figs. 5(a)–(c)), and line-scanning EDS also verified the uniform distribution of Ru NPs (Fig. 5(d)). Furthermore, the fast Fourier transformation (FFT) analysis of TEM images with the lattice plane intensity profile revealed that the interplanar spacing of metallic Ru (101) was 0.204 nm (Figs. 5(e) and (f)). On the other hand, it was found that the average size of Ru NPs increased to 3.1 nm in Ru/UiO-66 (Fig. S8).

XPS survey of Ru/UiO-66 and Ru/UiO-66_{def}-SO₃H found that C 1s spectra of both catalysts showed the carbon peak of C–O (289.2 eV), while additional C–S (287.7 eV) appeared in Ru/UiO-66_{def}-SO₃H (Fig. S9(a) and (b)). The overlapped Ru 3d signals were also observed in the range of 280–286 eV (Fig. S9(b)), confirming the introduction of Ru [46,47]. The O 1s spectra of two catalysts displayed three types of oxygen peaks in terms of O–H and/or S–O (533.1 eV), C–O and/or S=O (531.7 eV), and Zr–O (530.3 eV, Fig. S9(c)) [33]. Besides, the existence of S could be confirmed by S 2p_{1/2} (169.4 eV) and S 2p_{3/2} (168.2 eV) in Ru/UiO-66_{def}-SO₃H, which was absent in Ru/UiO-66 (Fig. S9(d)) [33]. Additionally, it was interesting to find that the binding energies of Zr 3d_{3/2} and Zr 3d_{5/2} located at 184.9 and 182.6 eV in Ru/UiO-66, and negatively shifted to 184.5 and 182.2 eV in Ru/UiO-66_{def}-SO₃H (Fig. S10) [36], which might be accredited to the electrons transfer from Ru NPs to unsaturated Zr–O nodes in defective clusters.

The binding energy of Ru 3p in Ru/UiO-66 could be deconvoluted into two pairs of double peaks, the pair peaks at 462.1 eV (3p_{3/2}) and 484.2 eV (3p_{1/2}) were assigned to Ru⁰ (metallic Ru), while the other at 465.1 eV (3p_{3/2}) and 487.8 eV (3p_{1/2}) might be attributed to Ru^{δ+} (RuO_x) in Fig. 4(b) [22,48]. Significantly, the pair peaks of Ru⁰ increased to 462.7 eV (3p_{3/2}) and 484.8 eV (3p_{1/2}), meanwhile, the other assigned to Ru^{δ+} shifted positively to 465.7 eV (3p_{3/2}) and 488.4 eV (3p_{1/2}) in Ru/UiO-66_{def}-SO₃H. The positive shift testified that more Ru species might be anchored in Zr–O nodes to form Ru–O–Zr and/or the electrons transferred from Ru to Zr–O in Ru/UiO-66_{def}-SO₃H than Ru/UiO-66 [48]. Besides, surface Ru content of Ru/UiO-66_{def}-SO₃H (4.69%) was lower than that of Ru/UiO-66 (6.13%). These results might be attributed to that

the size of Ru NPs in UiO-66_{def}-SO₃H (2.5 nm) was smaller than that of Ru/UiO-66 (3.1 nm, Table 1) and/or more Ru NPs would locate in mesoporous channels of Ru/UiO-66_{def}-SO₃H (Fig. S2(a)) as Ru NPs contacted strongly with defective sulfonated UiO-66_{def}-SO₃H than pristine UiO-66 [22].

3.2. Catalytic performance

Scheme S4 showed the successive pathway for the coupled methanolysis (of PET to DMT) and hydrogenation (of DMT to DMCD). At first, 2-hydroxyethyl methyl terephthalate (MHET) could be detected as an intermediate in the methanolysis of PET. And then, DMT formed after the further methanolysis of MHET [21]. Afterwards, DMCD was the target product *via* the

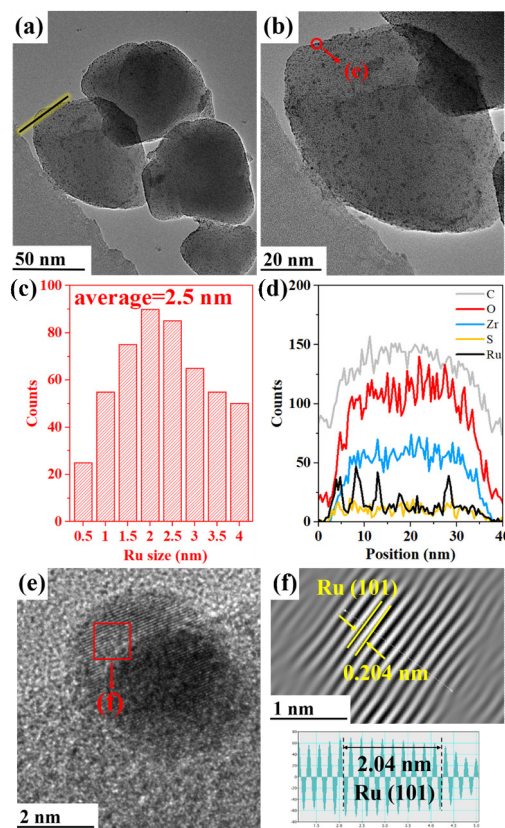


Fig. 5. TEM images (a,b,e), distributions of Ru NPs (c), line-scanning EDS (d), and FFT of TEM image with the lattice plane intensity profile (f) of Ru/UiO-66_{def}-SO₃H.

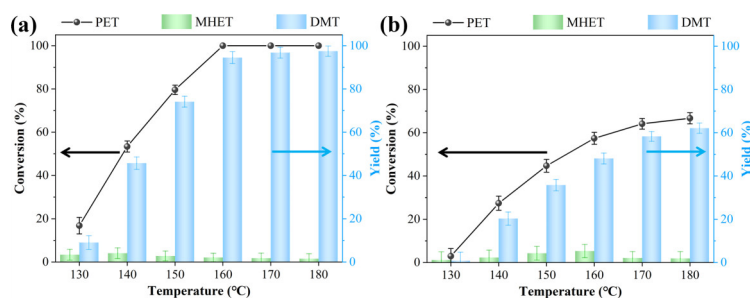


Fig. 6. Separated methanolysis of PET under varied reaction temperatures over Ru/Uio-66_{def}-SO₃H (a) and Ru/Uio-66 (b). Reaction conditions: 0.1 g catalyst, 2.0 g PET, 20 mL methanol, 2 h.

hydrogenation of benzene ring in DMT, with a byproduct, methyl 4-methylbenzoate (MMB), might form *via* the hydrogenation of an ester group [24].

Fig. 6(a) exhibited the performance of Ru/Uio-66_{def}-SO₃H for the separated methanolysis of PET (without the addition of H₂) under varied reaction temperatures. It was found that the conversion of PET increased quickly from 16.8% to 100% as the temperature raised from 130 to 160 °C (the TOF of each acid site reached 15.1 h⁻¹ at 130 °C), with the yield of DMT also increased sharply from 16.8% to 94.5%. Further increasing the temperature to 170 and 180 °C, the conversion of PET remained 100% and the yield of DMT enhanced slightly to 97%. Meanwhile, increasing temperature could reduce the yield of MHET (from 3.4% at 130 °C to 1.5% at 180 °C) and promote its further methanolysis to DMT. Compared to that of Ru/Uio-66_{def}-SO₃H, the catalytic activity of Ru/Uio-66 was suppressed under the same condition in Fig. 6(b). When the reaction temperature was raised from 130 to 180 °C, the conversion of PET merely grew from 2.9% to 66.7% (the TOF of each acid site was only 5.2 h⁻¹ at 130 °C), while the yield of DMT rose from 0.8% to 62.1%. The yield of MHET increased at first from 1.2% (130 °C) to its maximum of 5.3% at 160 °C, and then decreased to 1.9% (180 °C).

The performance of Ru/Uio-66_{def}-SO₃H for separated methanolysis of PET under varied reaction times (Fig. S11), different ratios between methanol and PET (Fig. S12) and catalyst dosages (Fig. S13) was described and discussed in Supporting Information. These results confirmed that Ru/Uio-66_{def}-SO₃H was active for the methanolysis of PET to DMT, and the highest yield of DMT reached 96.8% over 170 °C and 2 h.

The performance of Uio-66_{def}-SO₃H, Ru/Uio-66_{def}-SO₃H,

Uio-66, Ru/Uio-66 and without catalyst for the methanolysis of PET to DMT at 150 °C for 2 h was compared in Table S5. It was found that the introduction of -SO₃H could enhance the yield of DMT obviously due to the increased acidity (see Table S4). While the conversion of PET and yield of DMT decreased slightly after the loading of Ru NPs, which might be attributed to that some acid sites and pores were blocked by the introduction of Ru. And the degradation rate of PET without catalyst was less than 1%, indicating the essential of catalysts.

Fig. 7(a) displayed the performance of Ru/Uio-66_{def}-SO₃H for the separated hydrogenation of DMT under varied reaction temperatures. The conversion of DMT and yield of DMCD increased promptly from 20.2% to 97.4% and 18.5% to 95.7%, respectively, when the temperature was raised from 120 to 160 °C (the TOF of each Ru reached 47.9 h⁻¹ at 120 °C). As the temperature further increased to 170 °C, the conversion of DMT remained ~98% with high yield of DMCD (96.5%). And, the yield of MMB was below 1.5% at all temperatures. On the other hand, the performance of Ru/Uio-66 as catalyst was poor as the conversion of DMT just grew from 8.5% to 81.6% with the temperature rising from 120 to 170 °C (the TOF of each Ru was only 24.7 h⁻¹ at 120 °C), and the highest yield of DMCD only reached 79.3% at 170 °C in Fig. 7(b). Simultaneously, the yield of MMB was higher than 2.2% in these experiments.

The performance of Ru/Uio-66_{def}-SO₃H for separated hydrogenation of DMT to DMCD under varied reaction times (Fig. S14), hydrogen pressures (Fig. S15) and catalyst dosages (Fig. S16) was presented in Supporting Information. It was confirmed that Ru/Uio-66_{def}-SO₃H was also effective for the hydrogenation of DMT to DMCD, and the best yield of DMCD reached 96.5% under 170 °C and 3 MPa H₂ within 2 h.

To verify the performance of Ru/Uio-66_{def}-SO₃H for the

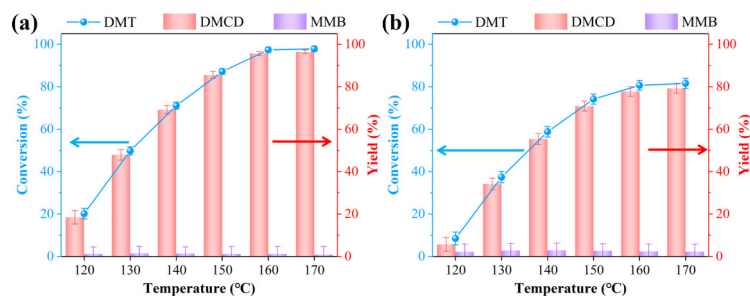


Fig. 7. Separated hydrogenation of DMT under varied reaction temperatures over Ru/Uio-66_{def}-SO₃H (a) and Ru/Uio-66 (b). Reaction conditions: 0.1 g catalyst, 2.0 g DMT, 20 mL methanol, 3 MPa H₂, 2 h.

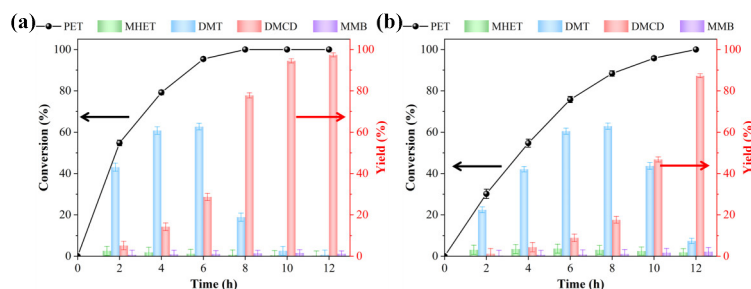


Fig. 8. Time course of direct upgrading of PET over Ru/UiO-66-def-SO₃H (a) and Ru/UiO-66 (b). Reaction conditions: 0.1 g catalyst, 2.0 g PET, 20 mL methanol, 160 °C, 3 MPa H₂.

coupled methanolysis and hydrogenation in one pot, the reaction mixture and catalyst after first-step methanolysis of PET at 160 °C for 2 h was directly used as the feedstock of second-step hydrogenation for another 2 h at 160 °C (Table S6). It was found that PET converted completely and the yield of DMCD reached 94.3%, which was close to separated methanolysis and hydrogenation (Figs. 6(a) and 7(a)). On the other hand, the conversion of PET and yield of DMCD were just 89.2% and 11.9% over Ru/UiO-66.

Encouraged by the excellent performance in above one-pot consecutive methanolysis and hydrogenation, Ru/UiO-66-def-SO₃H was utilized for direct upgrading of PET to DMCD at 160 °C and 3 MPa H₂, and the results were shown in Fig. 8(a). It was found that DMT was the main intermediate at the beginning of the reaction (0–6 h) and the yield of DMCD increased gradually, indicating the stepwise conversion of PET to DMCD *via* DMT. When the reaction time was prolonged to 8 h, the yield of DMCD reached 77.8% (with the yield of DMT decreasing to 18.9%) and PET converted completely. Moreover, the yield of DMCD further increased to 97.4% accompanied with 0.7%, 0.3% and 1.2% content of DMT, MHET and MMB as the reaction time was extended to 12 h. In comparison, the yield of DMCD was only 17.6% with 62.9% amount of DMT under 88.4% conversion of PET within 8 h over Ru/UiO-66 in Fig. 8(b).

In order to shorten the reaction time, direct upgrading of PET to DMCD was performed at 170 °C and these results were shown in Fig. 9(a). It was quite interesting to note that the conversion of PET reached 78.7% within 2 h, and the yield of DMT and DMCD enhanced to 63.8% and 10.4%, respectively. When the reaction time was prolonged to 4 and 5 h, the yield of DMCD increased quickly to 56.5% and 96.2% with the yield of

DMT dropping to 39.7% and 1.3% under complete conversion of PET. Moreover, the yield of DMCD further reached 97.7% accompanied with 0.9% content of DMT when the reaction time was extended to 6 h. These results indicated that increasing the temperature from 160 to 170 °C could effectively accelerate direct upgrading from PET to DMCD.

The reusability of Ru/UiO-66-def-SO₃H in direct upgrading of PET to DMCD at 170 °C within 4 h was manifested in Fig. 9(b). It was of great significance to note that Ru/UiO-66-def-SO₃H was quite stable in five recycles. The detected conversion of PET remained 100% in all cycles, and the yield of DMCD decreased slightly from 56.5% (in first usage) to 53.5% (in fifth usage), while the yield of DMT increased slightly from 39.7% to 42.5%.

Spent Ru/UiO-66-def-SO₃H was characterized and compared to the fresh one (Figs. S17–S20 and Tables S7, S8). It was found that the spent catalyst remained the UiO-66 skeleton with defects and cell parameters, and the characteristic peaks of metallic Ru still disappeared (Fig. S17 and Table S7). Furthermore, the spent Ru/UiO-66-def-SO₃H also showed type IV isotherms with a hysteresis loop at relative pressure ranging from 0.5 to 0.9, and the pore size distributions were similar with that of fresh one (Fig. S18 and Table S8). Besides, the chemical states of Ru species were similar as that of fresh catalyst, while Ru NPs kept well-anchored without obvious leaching (Fig. S19 and Table S7). And, the spent Ru/UiO-66-def-SO₃H maintained the morphology of regular octahedron resembled the fresh one, and there was no obvious agglomeration of Ru NPs (Fig. S20). These results indicated that the mesoporous structure and uniformly dispersed Ru NPs in Ru/UiO-66-def-SO₃H remained stable during the reaction.

At the same time, waste beverage bottles, packaging film and textile fiber were also used as feedstocks (Fig. S21) for

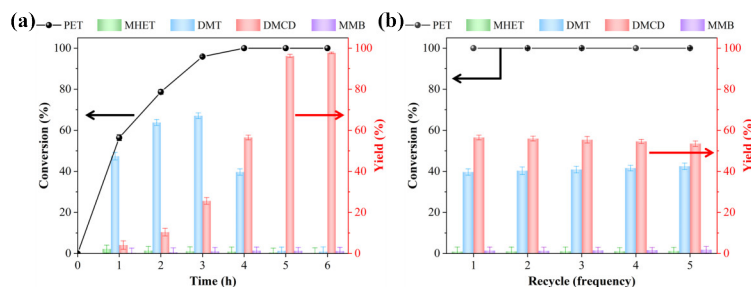


Fig. 9. Time course (a) and recycle usage (b) of direct upgrading of PET over Ru/UiO-66-def-SO₃H. Reaction conditions: (a) 0.1 g catalyst, 2.0 g PET, 20 mL methanol, 170 °C, 3 MPa H₂; (b) 0.1 g catalyst, 2.0 g PET, 20 mL methanol, 170 °C, 3 MPa H₂, 4 h.

direct upgrading of PET to DMCD over Ru/Uio-66_{def}-SO₃H, and these results were summarized in Table 3. The yields of DMCD were higher than 92% with complete conversions of PET at 170 °C and 3 MPa H₂ within 6 h for three PET-based products, which indicated that Ru/Uio-66_{def}-SO₃H had potential in the application for the upcycling of various waste PET-based feedstocks.

3.3. Catalyst structure-performance relationship

Direct formation of DMCD *via* waste PET can avoid the difficult and expensive separation process of DMT from solid catalyst and/or remaining PET, but the efficient catalyst should be versatile as it must contain both acid sites (for the methanolysis of PET) and Ru NPs (for the hydrogenation of DMT, see Scheme S4). In this work, the special designed Ru/Uio-66_{def}-SO₃H possesses abundant acidity (580 μmol·g⁻¹, see Table S4), mesoporous channels (2.72 and 3.44 nm), high surface area and pore volume (998 m²·g⁻¹ and 0.83 cm³·g⁻¹, Fig. 2 and Table 2). These properties make Ru/Uio-66_{def}-SO₃H exhibit excellent performance for the first-step methanolysis of PET to DMT. The calculated TOF of each acid site in Ru/Uio-66_{def}-SO₃H reached 15.1 h⁻¹ at 130 °C, and this activity was higher than that of Ru/Uio-66 (only 5.2 h⁻¹ at 130 °C, Fig. 6). At the same time, high hydrophilicity and strong acidity of -SO₃H can also make the methanolysis of PET to DMT over Ru/Uio-66_{def}-SO₃H catalyst accelerated among those reported solid catalyst including Cu/SiO₂-IM, Cu/ZrO₂, Ti_{0.5}Si_{0.5}O₂, calcined Na₂SiO₃ and MgO/NaY (Table S9).

Besides, unsaturated Zr–O nodes and –SO₃H can anchor Ru, and the dispersion of Ru in Ru/Uio-66_{def}-SO₃H reaches 45.1% (see Table 1), which is beneficial for the adsorption (108.8 μmol·g⁻¹) and activation of hydrogen (Fig. S6), further accelerates the hydrogenation of DMT [22]. The positive shift of Ru species in Ru/Uio-66_{def}-SO₃H also testifies the active electron interactions between Ru to Zr–O (Fig. 4(b)), enhancing the hydrogenation ability of Ru. Meanwhile, the mesoporous channels (2.72 and 3.44 nm) and enlarged pore volume (0.83 cm³·g⁻¹) in Ru/Uio-66_{def}-SO₃H can promote the diffusion of substance and the contact between reactants and Ru NPs (Fig. 2, Table 2 and Scheme S2) [43]. In addition, sufficient acidity makes the benzene ring absorbed *via* “non-coplanar” on Ru species and promote the selective hydrogenation of benzene ring (see Table S4) [49,50]. The calculated TOF of Ru in Ru/Uio-66_{def}-SO₃H reached 47.9 h⁻¹ at 120 °C, which was higher than that of Ru/Uio-66 (only 24.7 h⁻¹ at 120 °C, Fig. 7). These properties make Ru/Uio-66_{def}-SO₃H show higher performance in the hydrogenation of DMT to DMCD compared to those reference

Ru-based catalysts including Ru₄Pt₂/Al₂O₃, Ru_{0.4}Ni₂/CNT, Ru/Ni/Ni(Al)O_x, Ru/SBA-15 and Ru/HZSM-5 (see Table S10).

On the other hand, the reaction energy barrier diagram for the hydrogenation of benzene ring (DMT) to cyclohexyl (DMCD) over Ru was displayed in Fig. S22 [51,52]. As the relative energy of cyclohexyl (–4.78 eV) was lower than that of benzene ring (0.00 eV), density functional theory (DFT) calculations suggested that Ru cluster was efficient for the hydrogenation of benzene ring (DMT) to cyclohexyl (DMCD) *via* sequential hydrogenation following the Horiuti-Polanyi type mechanism (Fig. S23 and Table S11) [53,54].

These characteristics of Ru/Uio-66_{def}-SO₃H indicates that Ru/Uio-66_{def}-SO₃H simultaneously possesses enlarged pore channels (mesopores and high pore volume), abundant acidity and hydrophilicity (unsaturated Zr–O nodes and –SO₃H), sufficient hydrogenation active sites (highly dispersed Ru NPs), which confirms the versatility and high efficiency for the methanolysis of PET (Fig. 6(a)) and hydrogenation of DMT (Fig. 7(a)), realizing direct upgrading of PET to DMCD in one pot (Figs. 8(a) and 9(a)).

Furthermore, Ru/Uio-66_{def}-SO₃H also possesses the outstanding chemical and thermal stability (Fig. 9(b)), which can be ascribed to abundant acid sites (see Table S4), strong interaction between Ru species and Uio-66_{def}-SO₃H (Fig. S19), as well as mesoporous channels to support Ru NPs (Fig. S18 and Table S8), avoiding the aggregation and leaching of Ru NPs during the reaction (Fig. S17, Fig. S20 and Table S7).

That is, the special designed Ru/Uio-66_{def}-SO₃H integrating the active sites for the methanolysis of PET and hydrogenation of DMT achieves direct upgrading of PET to DMCD in one pot with excellent activity and stability.

3.4. Possible reaction kinetics and mechanism

According to above experiments, the possible reaction kinetics of methanolysis of PET, hydrogenation of DMT, direct upgrading of PET to DMCD over Ru/Uio-66_{def}-SO₃H was analyzed to further reveal the reaction mechanism. Based on the separated performance of methanolysis of PET to DMT over Ru/Uio-66_{def}-SO₃H (Fig. S11), it was found that the methanolysis of PET was approximate first-order reaction of remaining concentration of PET (Fig. S24) [55,56], and the calculated apparent rate constant (at 150 °C) was about 0.7667 h⁻¹. On the other hand, the rate of separated hydrogenation of DMT over Ru/Uio-66_{def}-SO₃H (Fig. S14) was unattached with the remaining concentration of DMT in the reaction mixture (Fig. S25), and the calculated apparent rate constant (at 130 °C) was about 0.1192 mol_{DMT}·L⁻¹·h⁻¹.

In the direct upgrading of PET to DMCD over Ru/Uio-66_{def}-SO₃H, the calculated rate constant of consuming PET at 160 °C was 0.4926 h⁻¹ (Figs. 8(a) and S26), and it increased obviously to 0.7732 h⁻¹ at 170 °C (Figs. 9(a) and S27). These results indicated that the direct upgrading of PET to DMCD over Ru/Uio-66_{def}-SO₃H was difficult, and the consuming rate of PET depended strongly on reaction temperature.

At the same time, it was found that the selectivity of MHET (intermediate product of DMT, see Scheme S4) decreased con-

Table 3

Direct upgrading of varied PET-based products over Ru/Uio-66_{def}-SO₃H.

Entry	Substrate	Conversion of PET (%)	Yield of DMCD (%)
1	beverage bottles	100	95.4
2	textile fiber	100	93.2
3	packaging film	100	92.8

Reaction conditions: 0.1 g catalyst, 2.0 g feedstock, 20 mL methanol, 170 °C, 3 MPa H₂, 6 h.

tinuously from 3.9% (1 h) to 0.2% (6 h) during the direct conversion of PET to DMCD (Fig. 9(a)), indicating that MHET formed at the beginning of the reaction and could further convert to DMT (Scheme S4 and Fig. S11). Moreover, the selectivity of DMT first rose to 81.1% (2 h) and then declined to 0.9% (6 h) with the increasing formation of DMCD. These results indicated that the formation DMT from PET and/or MHET proceeded quickly than the followed hydrogenation of DMT to DMCD. In addition, the selectivity of MMB (the byproduct of the hydrogenation of ester group in DMT) increased slightly to 1.4% (6 h) due to the methylation under long reaction time.

According to the above results and related literatures, we think that the overall reaction paths over Ru/UiO-66_{def}-SO₃H should be carried out in the following steps. At first, acid sites (including Zr–O nodes, Zr–OH and –SO₃H) on Ru/UiO-66_{def}-SO₃H interacted with the carbonyl oxygen in PET, while the carbonyl carbon was attacked by the nucleophilic hydroxyl in methanol, resulting in chain cleavage and modification of the ester group. In this process, DMT was the main product *via* methanol transesterification on both sides, and there was also small amount of MHET formed *via* methanol transesterification at one side (or by transesterification of ethylene glycol with DMT [57]), which could further convert to DMT. Then, the delocalized π conjugated band of benzene ring in DMT was activated on Ru sites, and further the activated hydrogen, which was adsorbed and dissociated on Ru, attacked the delocalized π conjugated band, and the benzene ring was gradually hydrogenated to generate the cyclohexyl (DMCD). Due to the methylation, part of DMCD was converted into MMB especially at high reaction temperature and H₂ pressure for long reaction time.

4. Conclusions

In summary, defective sulfonated UiO-66_{def}-SO₃H was synthesized *via* defect engineering and sulfonation of UiO-66, and Ru/UiO-66_{def}-SO₃H catalyst could be prepared *via* impregnation of RuCl₃ and reduction. Characterization indicated that the mesoporous Ru/UiO-66_{def}-SO₃H possessed enlarged surface area and pore volume, enhanced acid sites, and high dispersion of Ru than those of Ru/UiO-66. As a result, Ru/UiO-66_{def}-SO₃H not only showed prominent performance in separated methanolysis of PET to DMT and hydrogenation of DMT to DMCD, but also processed potential for direct upgrading of PET to DMCD under mild condition. PET could be converted completely and the yield of DMCD reached 97.7% at 170 °C and 3 MPa H₂ within 6 h. It was concluded that Ru/UiO-66_{def}-SO₃H combining the newly emerging mesopores, abundant acidity (active sites of methanolysis), high dispersion of Ru (active sites of hydrogenation) and hydrophilic sites (adsorption of PET and its oligomers) was efficient for direct conversion of PET to DMCD. And, Ru/UiO-66_{def}-SO₃H remained stable after five cycles due to the strong interaction between Ru NPs and Zr–O nodes, and the mesopores for loading Ru NPs. Moreover, Ru/UiO-66_{def}-SO₃H was also versatile for the upgrading of varied PET-based products including beverage bottles, packaging film and textile fiber.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgments

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Electronic supporting information

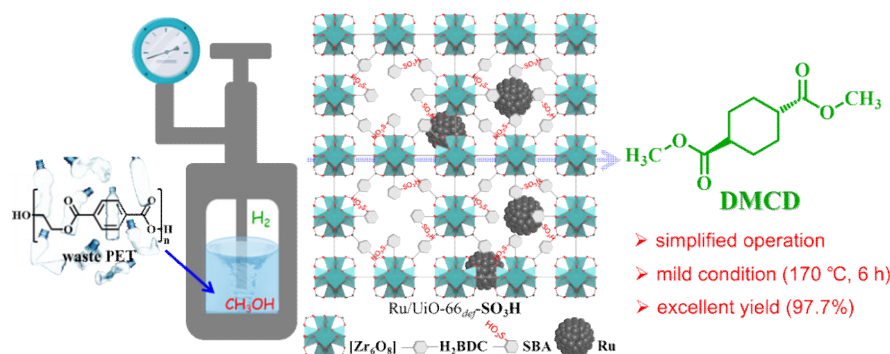
Supporting information is available in the online version of this article.

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Graphical Abstract

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Catalytic upgrading of waste PET to dimethyl cyclohexane-1,4-dicarboxylate over defective sulfonated UiO-66_{def}-SO₃H supported Ru catalystWeitao Ou, Yingdan Ye, Yibin Zhang, Huaiyuan Zhao *, Weichen Du, Zhaoyin Hou *
Zhejiang University; Zhejiang Hengyi Petrochemical Research Institute Co., Ltd.Catalytic upgrading of waste PET to DMCD over Ru/UiO-66_{def}-SO₃H

Defective sulfonated UiO-66_{def}-SO₃H supported Ru catalyst was efficient in the direct upgrading of waste polyethylene terephthalate (PET) to dimethyl cyclohexane-1,4-dicarboxylate (DMCD), and the yield of DMCD reached 97.7% at 170 °C within 6 h.

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废弃PET化学升级为DMCD的缺陷型磺酸化Ru/UiO-66_{def}-SO₃H催化剂

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摘要: 对苯二甲酸-乙二醇-1,4-环己烷二甲醇共聚酯(PCT/PETG)以及对苯二甲酸-四甲基环丁二醇-环己二醇共聚酯(PCTG)具有耐热性高、韧性好、抗冲击、透明和可回收等优势, 广泛用于婴儿制品、儿童玩具、高端显示材料和高性能粉末涂料等领域。然而, 合成上述共聚酯的关键单体1,4-环己烷二甲醇(CHDM)严重依赖进口, 价格昂贵, 传统的对苯二甲酸二甲酯(DMT)两步加氢制备CHDM需要使用贵金属(Pd)催化剂, 工艺条件苛刻。直接采用废弃聚对苯二甲酸乙二醇酯(PET)为原料化学升级制备CHDM及其关键中间体1,4-环己烷二甲醇二甲酯(DMCD)可以取代昂贵的DMT原料, 研制可以实现这个过程的催化剂和反应机理具有十分重要的意义和应用前景, 这也是目前废弃聚酯转化方向的研究热点。

针对PET的链状结构、中间产物(DMT)的分子体积大(1.32–1.37 nm)、 “一锅”法化学升级制备DMCD需要醇解活性中心和加氢活性中心的协同作用以及适宜的孔径和表界面特性等。本文制备了缺陷型磺酸化UiO-66_{def}-SO₃H负载的Ru基催化剂(Ru/UiO-66_{def}-SO₃H), 成功地利用该催化剂实现了废弃PET “一锅”法化学升级制备DMCD的新工艺。表征结果显示, 与传统的微孔型Ru/UiO-66催化剂相比, Ru/UiO-66_{def}-SO₃H中生成了规则的介孔孔道(2.72和3.44 nm), 具有较大的比表面积和孔体积(分别为998 m²·g⁻¹和0.83 cm³·g⁻¹), 酸量大(580 μmol·g⁻¹), 亲/疏水性(接触角为22.9°)适宜, Ru与载体(Zr-O)相互作用较强, Ru分散度高(45.1%)等优势。Ru/UiO-66_{def}-SO₃H催化剂在单独的PET醇解到DMT, 以及DMT加氢到DMCD过程中都具有较好的活性, 这主要归因于配位不饱和Zr-O节点的暴露以及-SO₃H的引入可以增加酸性位点(PET醇解活性位点)的数量, 介孔孔道有利于DMT/DMCD等的扩散, 亲水性能提升能够增强带有末端羟基的PET及其低聚物的吸附, 而且Ru的分散度高, 与载体的强相互作用, 催化剂的酸性等也可以进一步促进DMT加氢反应的进行。上述特征也是废弃PET “一锅”法化学升级制备DMCD的关键。然后, 直接以PET为原料, 发现在 $m(\text{cat.})/m(\text{PET}) = 0.05 \text{ g} \cdot \text{g}^{-1}$, 170 °C, 3 MPa H₂条件下反应6 h, PET完全转化(100%), DMCD产率高达97.7%。并且, Ru/UiO-66_{def}-SO₃H在PET基废弃产品(饮料瓶, 纺织纤维, 包装膜)直接化学升级为DMCD反应中也具有较好的性能。最后, 通过详细表征反应液中的可溶产物和残余固体的组成, 并结合理论计算, 分析了PET醇解以及DMT加氢的反应路径和机理, 发现PET “一锅”法化学升级制备DMCD首先需要在催化剂的酸性中心作用下醇解解聚成中间体DMT, 然后DMT再进一步加氢。可以实现上述步骤有效衔接的关键是催化剂的酸性、加氢活性、介孔结构和表面亲疏水性的高效协同。

综上, 本工作中研制的缺陷型磺酸化Ru/UiO-66_{def}-SO₃H催化剂可以将酸性中心、加氢活性中心和介孔结构和表面亲疏水性等高效地组合在一起, 这些特性的协同作用弥补了各自的不足。该催化剂在单独的PET醇解、DMT加氢以及废弃PET “一锅”法化学升级制备DMCD反应中都具有优异的活性。本文为设计和合成高效催化剂以及废弃PET的化学升级转化提供了新思路。

关键词: 废弃聚对苯二甲酸乙二醇酯; 化学升级; 1,4-环己烷二甲醇二甲酯; Ru/UiO-66_{def}-SO₃H

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